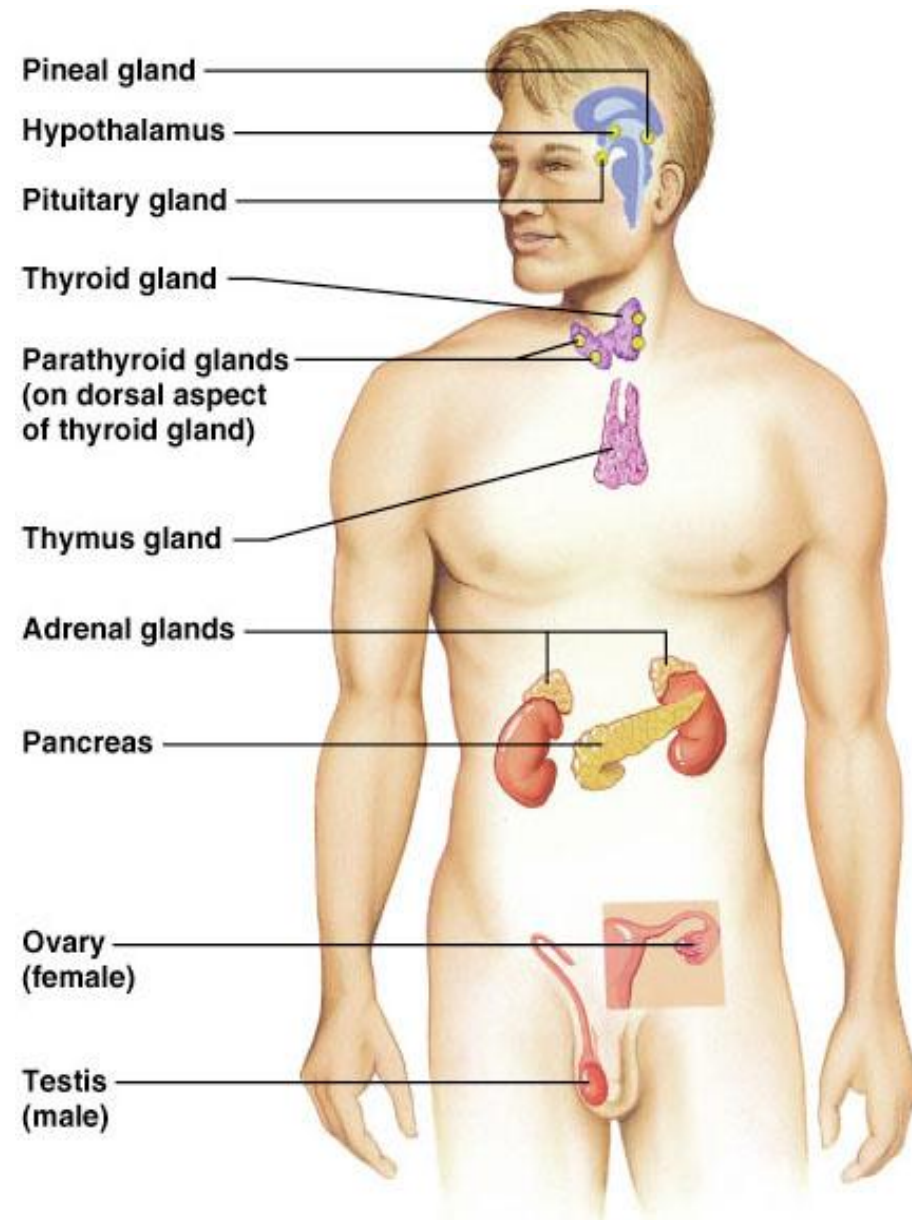

Physiology of the Endocrine System

Introduction

Endocrine Glands

- Hypothalamus
- Pituitary gland
- Pineal gland
- Thyroid gland
- Parathyroid glands
- Adrenal gland (cortex & medulla)
- Pancreas (Islets of Langerhans)
- Gonads (Testes & Ovary)
- Placenta...



Endocrine Glands

- Pineal gland → secrete Melatonin
 - Inhibit gonadal function and delays the onset of puberty
- Thymus gland → secrete Thymopoietin
 - Stimulate lymphocyte proliferation
- Heart → secrete ANP → cause Natriuresis
- Kidney → secrete Renin, erythropoietin...
- Liver → secrete Erythropoietin, Somatomedin-C...
- GIT → secrete Gastrin, Secretin, CCK, VIP, Sub-P...
-

Hormones

- Secreted by endocrine glands/cells into the blood
- Some are stored, others are manufactured as needed
- Transported to distant targets by binding to plasma proteins or dissolved in plasma
- Bind to their receptors on target organs and influence their function
 - Their effect on tissues depends on the type of receptors
- They regulate d/t physiological processes:
 - Reproduction + Metabolism
 - Growth and development + Homeostasis

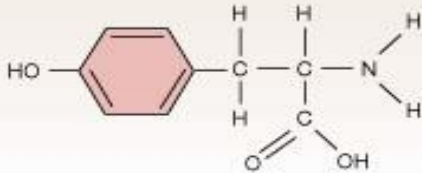
Hormones

A. Classification (Based on chemical structure)

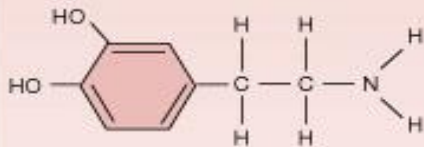
1. Amines:

- Derived from amino acids (tyrosine, tryptophan...)
- Adrenal medulla → EP, NE → derived from Tyrosine
- Thyroid gland → T3, T4 → derived from Tyrosine
- Pineal gland → Melatonin → derived from Tryptophan

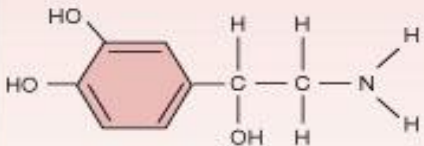
Tyrosine
is the parent amino acid for catecholamines and thyroid hormones.



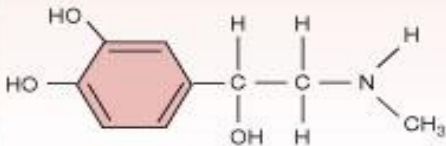
Catecholamines
are made by modifying the side groups of tyrosine.



Dopamine

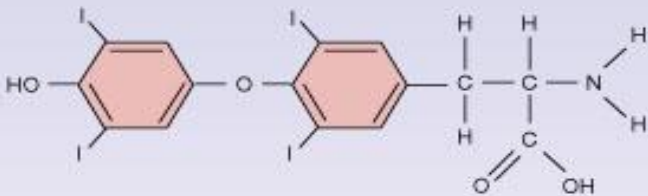


Norepinephrine

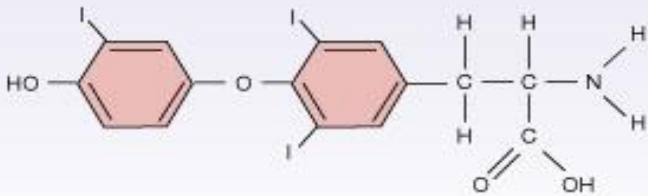


Epinephrine

Thyroid hormones
are synthesized from two tyrosines and iodine (I) atoms.



Thyroxine (Tetraiodothyronine, T₄)



Triiodothyronine (T₃)

Amine Hormones

Hormones

2. Peptides and proteins:

- Water soluble hormones synthesized by RER & stored in secretory vesicles until needed
- Pituitary hormones → GH, ACTH, Prolactin + ADH, Oxytocin
- Pancreas → Insulin & Glucagon
- Parathyroid Hormone

3. Glycoproteins

- Consist of long polypeptide chains (>100aa) bound to CHO groups
- FSH, LH, TSH

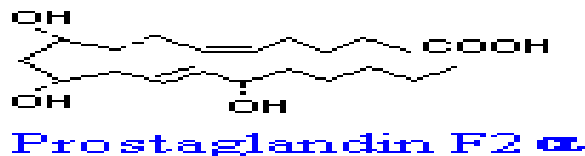
Hormones

4. Steroids

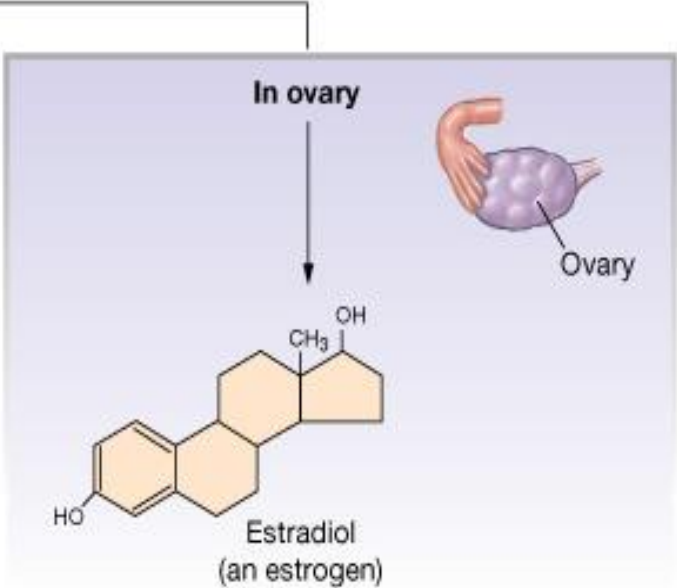
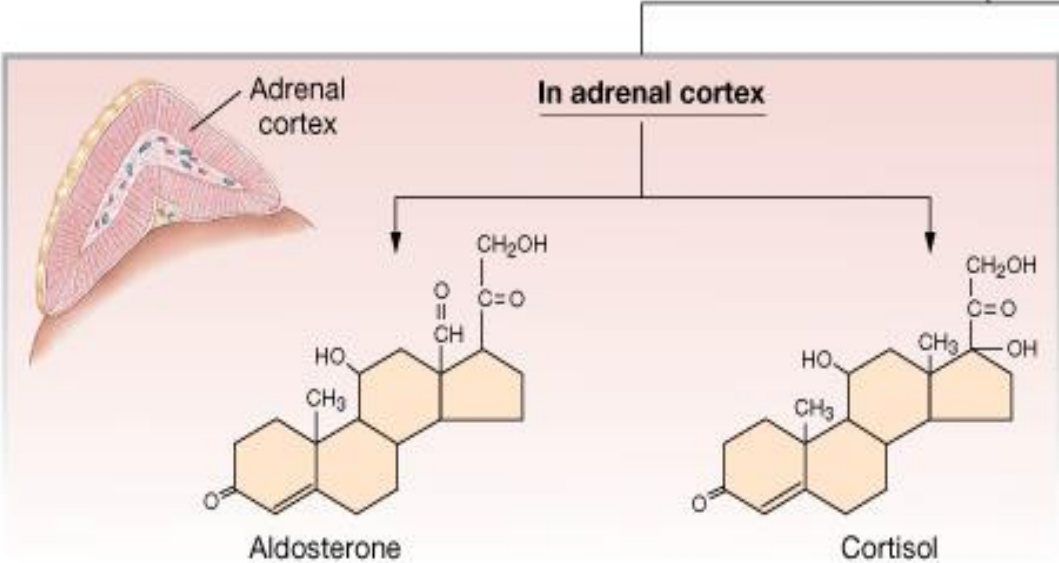
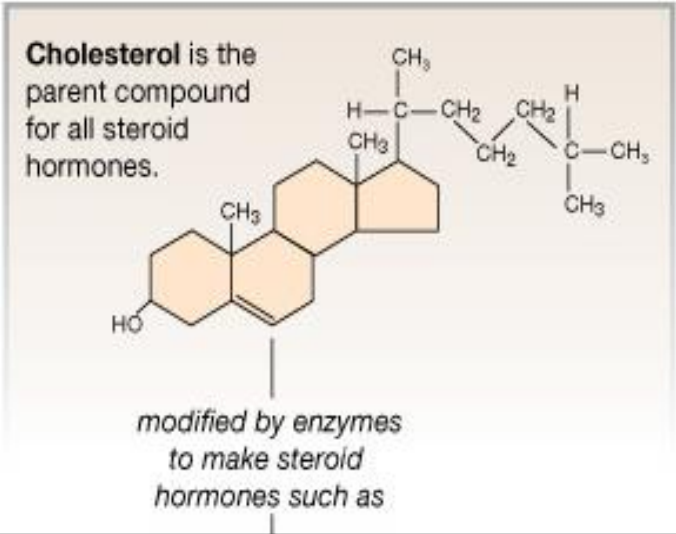
- Lipid hormones synthesized from cholesterol
 - *Usually synthesized when needed (not stored)*
- Adrenal cortex → corticosteroids + sex steroids (little)
- Gonads → testosterone, estradiol, progesterone

5. Eicosanoids

- Derived from polyunsaturated fatty acids
- Prostaglandin's + Leukotriene's + Thromboxane's



Steroid Hormones



Hormones

B. Based on water solubility

1. Hydrophilic hormones

- Polar and water soluble → can't cross the plasma membrane
- Catecholamines, peptide/protein hormones

2. Lipophilic hormones

- Non-polar and water insoluble (*lipid soluble*) → can cross the plasma membrane
- Thyroid and steroid hormones

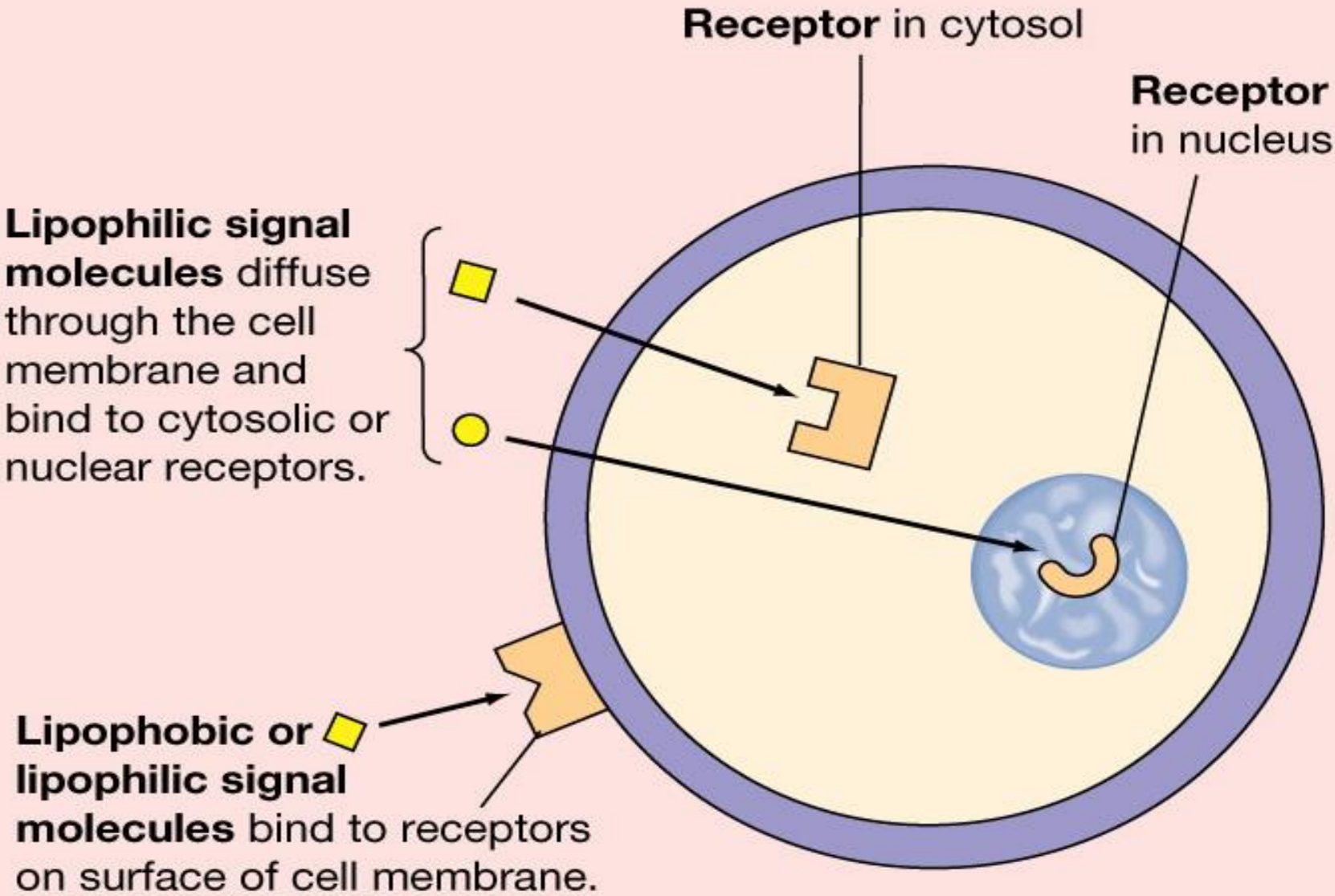
Hormone Receptors

1. Membrane receptors:

- Receptors for hydrophilic hormones
- Specific mostly for protein, peptide & catecholamines

2. Cytosolic/Nuclear receptors

- Receptors for steroids & thyroid hormones, Vitamin-D3
- Retinoid X receptor (RXR) → receptor for Retinoic acid (derived from vit-A(Retinol))
- Often activates genes → slower response



Hormone Receptors

- Receptor Upregulation
 - The stimulating hormone induces greater than normal formation of receptors or intracellular signaling molecules
- Receptor Downregulation (Desensitization)
 - Usually subsequent exposure to high concentrations of hormones decreases the number of receptors in target cells → decreases target cell response
 - Adipose cells exposed to high concentrations of insulin
 - Testicular cells exposed to high concentrations of LH

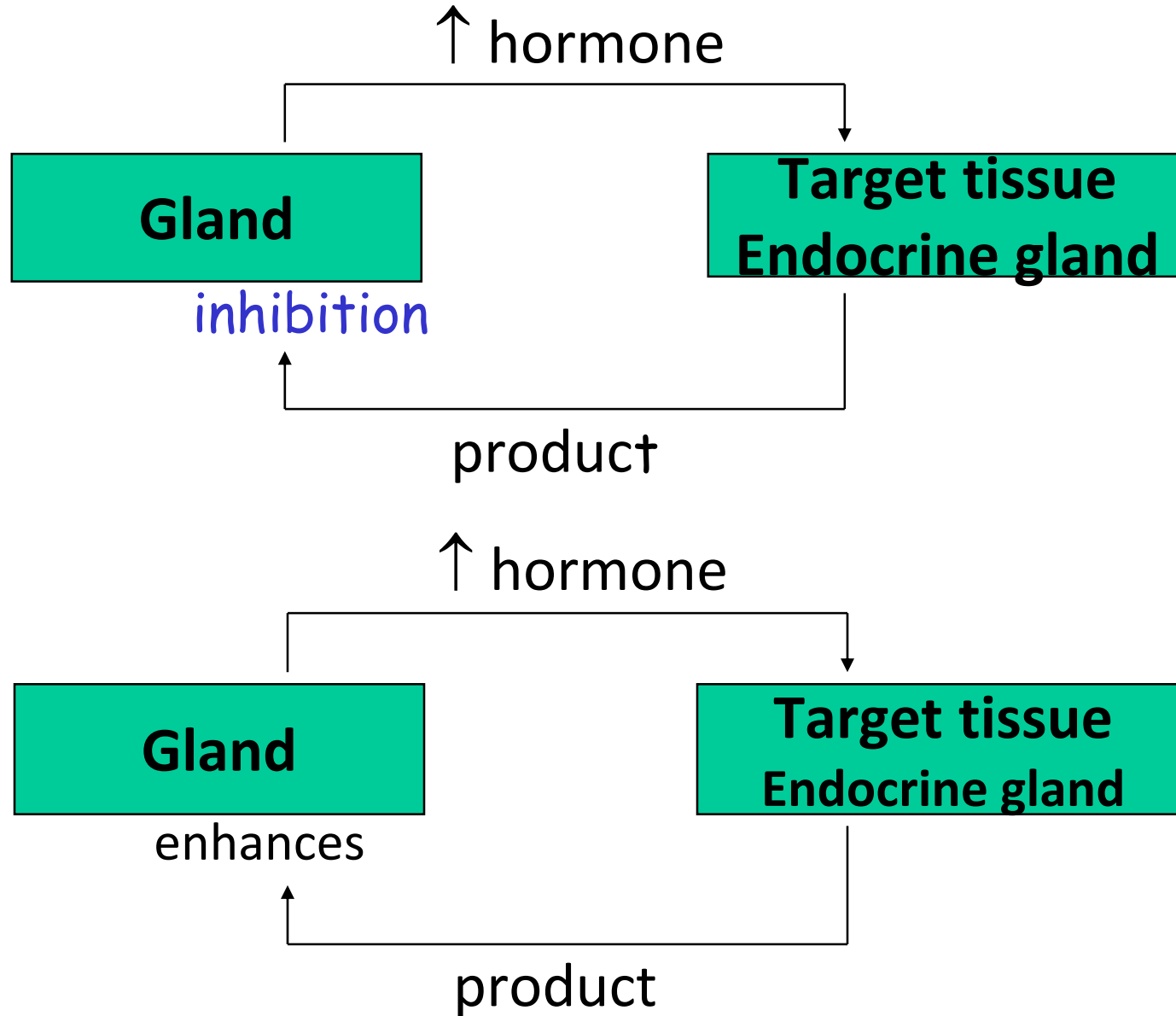
Hormone Receptors

- **Receptor Downregulation: causes**
 - Inactivation of receptors + intracellular signaling molecules
 - Temporary sequestration of the receptor to the inside of the cell
 - Destruction of receptors by lysosomes after they are internalized
 - Decreased production of receptors
- **NB**
 - **Pulsatile secretion** of hormones prevents desensitization (*Pulsatile secretion of GnRH and LH*)

Hormones

- Regulation of hormone secretion:
 1. Feedback regulatory mechanisms
 - Negative feedback mechanism (most hormones)
 - Positive feedback mechanism (LH- surge)
 2. Circadian rhythm (periodic variations)
 - Cortisol, sex hormones, GH

Negative & positive feedback mechanism



Hormone Interactions

- Multiple hormones can affect a single target simultaneously
- Hence, the response of a target tissue to a particular hormone is affected by the effects of other hormones on that tissue
- Hormone interactions can be:
 1. Synergistic (additive + complementary)
 2. Permissive
 3. Antagonistic

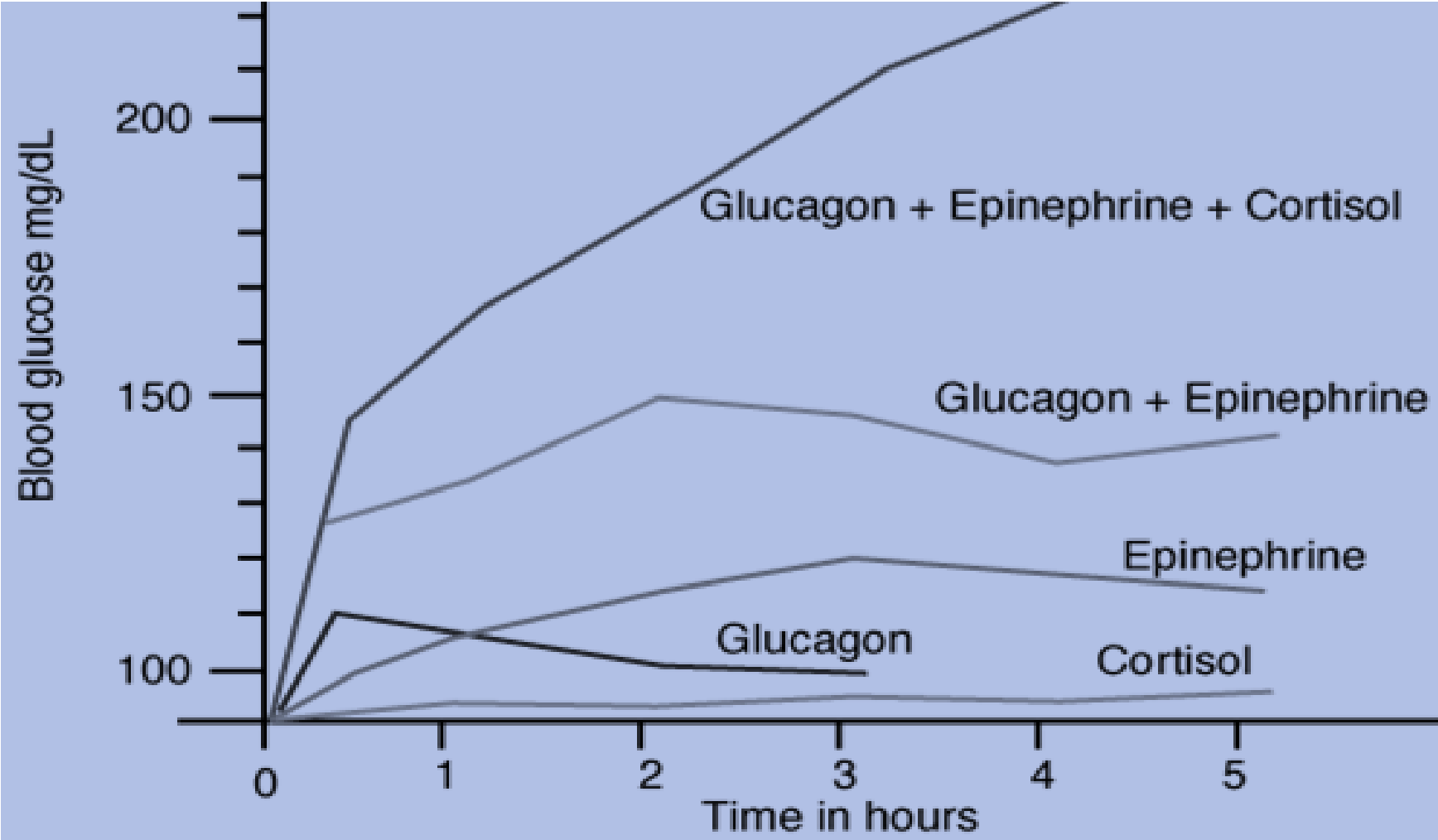
Hormone Interactions

1. Synergistic interaction

- Two or more hormones interact to produce more effect

a. Additive effect

- Each hormone separately activates the same physiological process on the target cell
- Combined effect of the hormones is higher than the effect caused by a single hormone
- EP and NE → increase heart rate
- Glucagon + EP + Cortisol → increase blood glucose level



Synergistic effects of Glucagon, Cortisol and EP on Blood Glucose

Hormone Interactions

b. Complementary effect

- Each hormone separately stimulates a different stage of the same physiological process and their combined effect is more than just additive
- **Estrogen + Prolactin + Oxytocin** → act on mammary glands for milk secretion

Hormone Interactions

2. Permissive Interaction

- One hormone enhances the response of a target organ to the second hormone or it increases the activity of the second hormone
- **Estrogen and Progesterone** → Prior exposure of the uterus to estrogen induces the formation of progesterone receptors → increase the response of uterus to progesterone
- **Cortisol and Epinephrine** → Cortisol increases responsiveness of vascular smooth muscles for catecholamines
- **PTH and Calcitriol** → PTH stimulates the synthesis of hydroxylating enzymes (*Liver + kidney*) for vitamin-D activation

Hormone Interactions

3. Antagonistic Interaction

- Each hormone have different effects on the target organ/physiological process
- Insulin & Glucagon → on blood glucose level
- Aldosterone & ANP → on sodium excretion
- PTH & Calcitonin → on serum calcium level

Hormones: MoA

- Generally, hormones of the same chemical category have similar mechanisms of action

1. Lipid-soluble hormones

- Pass through the target cell's plasma membrane, bind to intracellular receptors, and act directly within the target cells

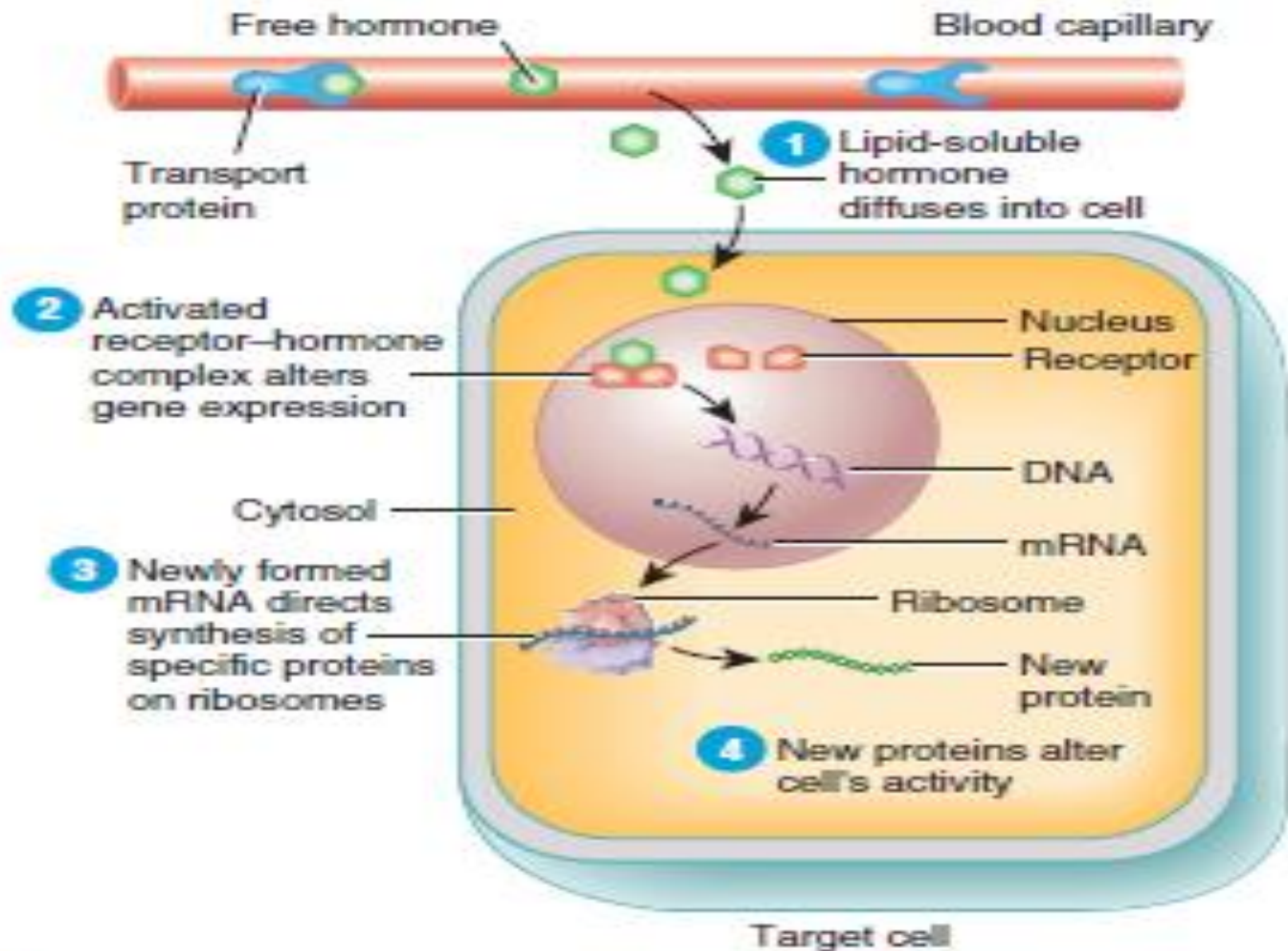
2. Water-soluble hormones

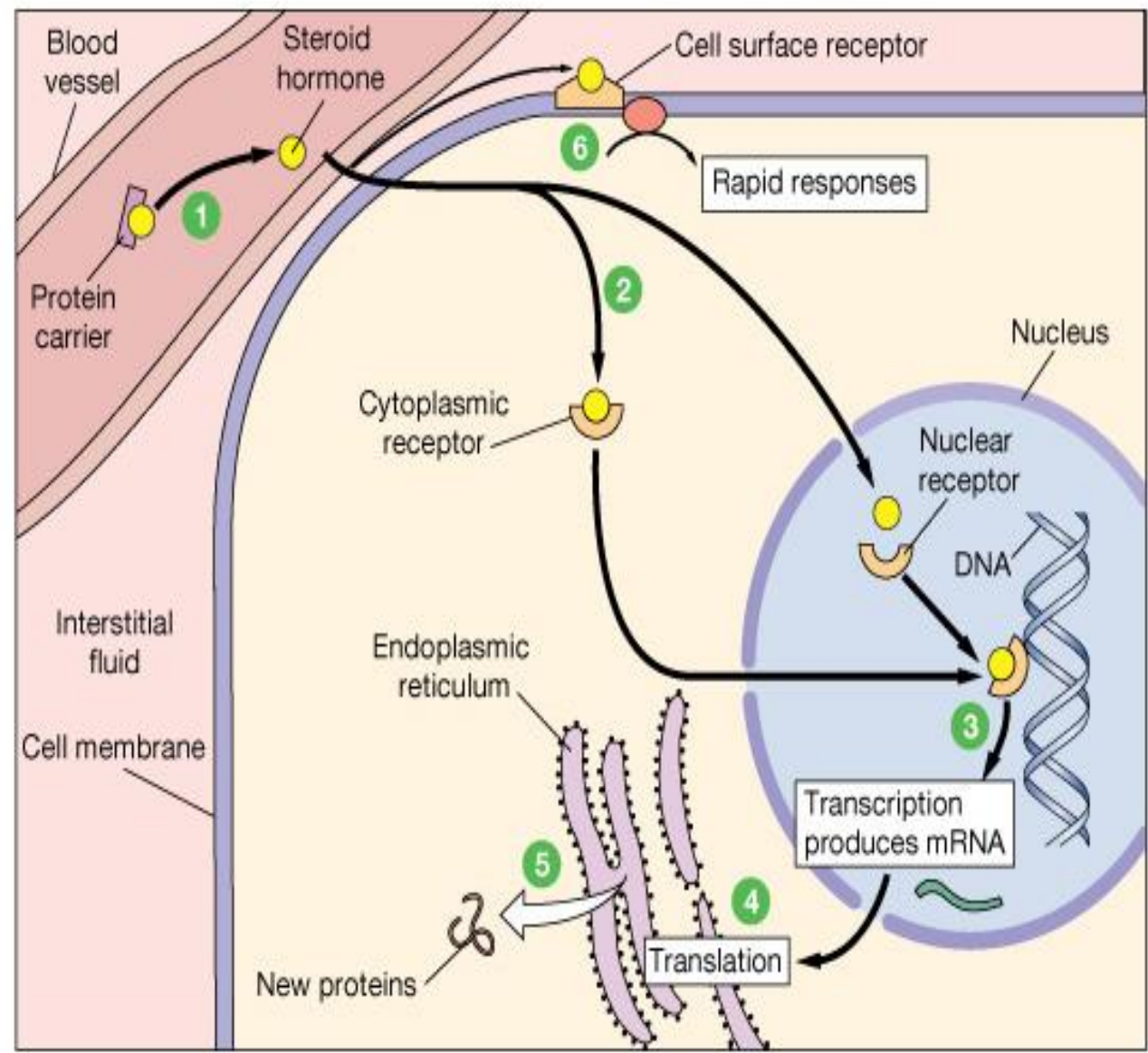
- Bind to membrane receptors → activate intracellular second-messenger systems that mediate the actions of the hormone

Hormones: MoA

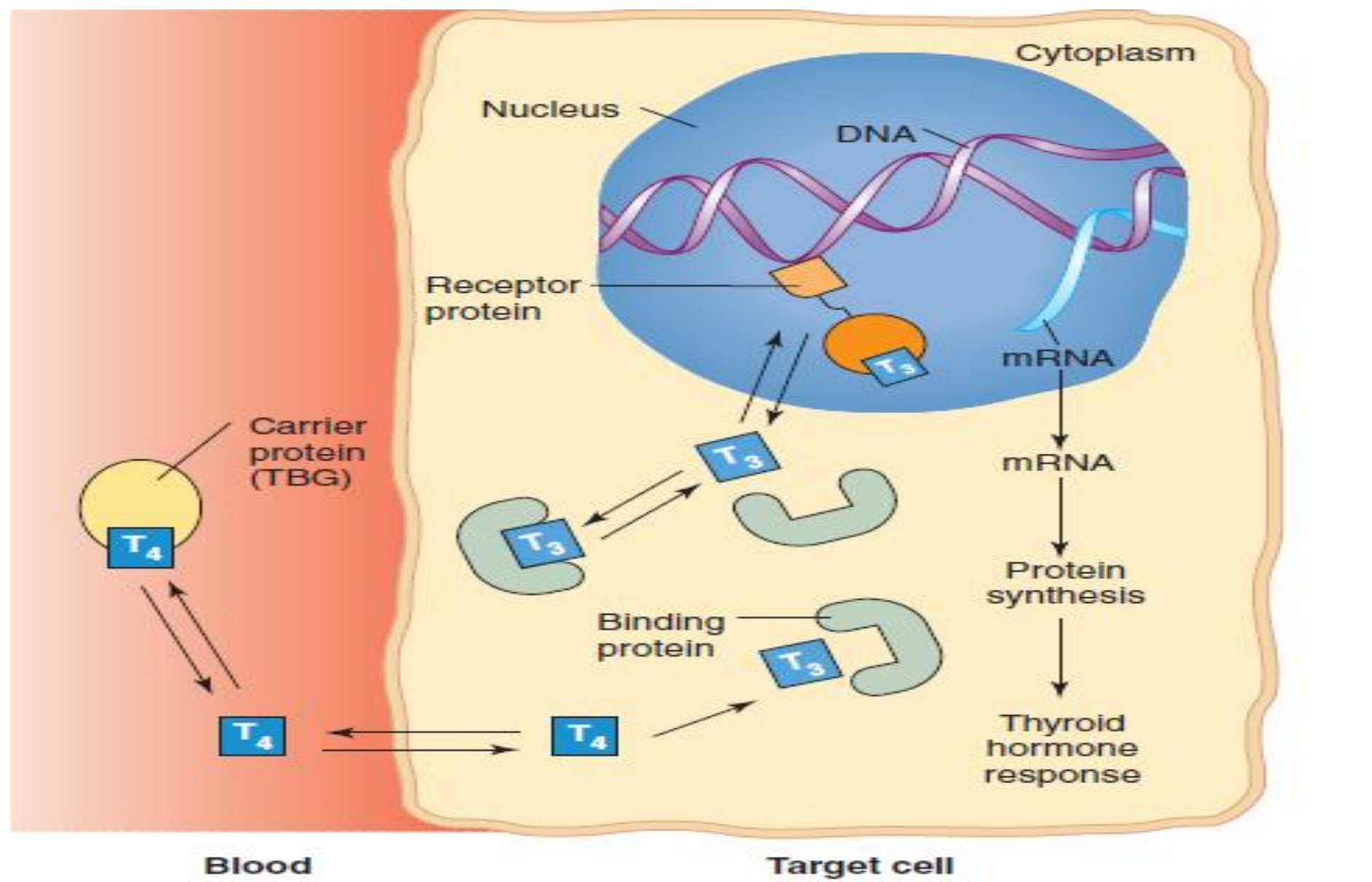
1. Lipophilic hormones

- Bind to nuclear receptors and activate/inhibit DNA transcription
- Nuclear receptors have two domains:
 - A ligand (hormone) binding domain and
 - A DNA-binding domain (hormone-response element)
- The newly formed mRNA directs the synthesis of specific proteins (enzymes...) that alter the target cell's activity





- 1 Most hydrophobic steroids are bound to plasma protein carriers. Only unbound hormones can diffuse into the target cell.
- 2 Steroid hormone receptors are in the cytoplasm or nucleus.
- 3 The receptor-hormone complex binds to DNA and activates or represses one or more genes.
- 4 Activated genes create new mRNA that moves back to the cytoplasm.
- 5 Translation produces new proteins for cell processes.
- 6 Some steroid hormones also bind to membrane receptors that use second messenger systems to create rapid cellular responses.

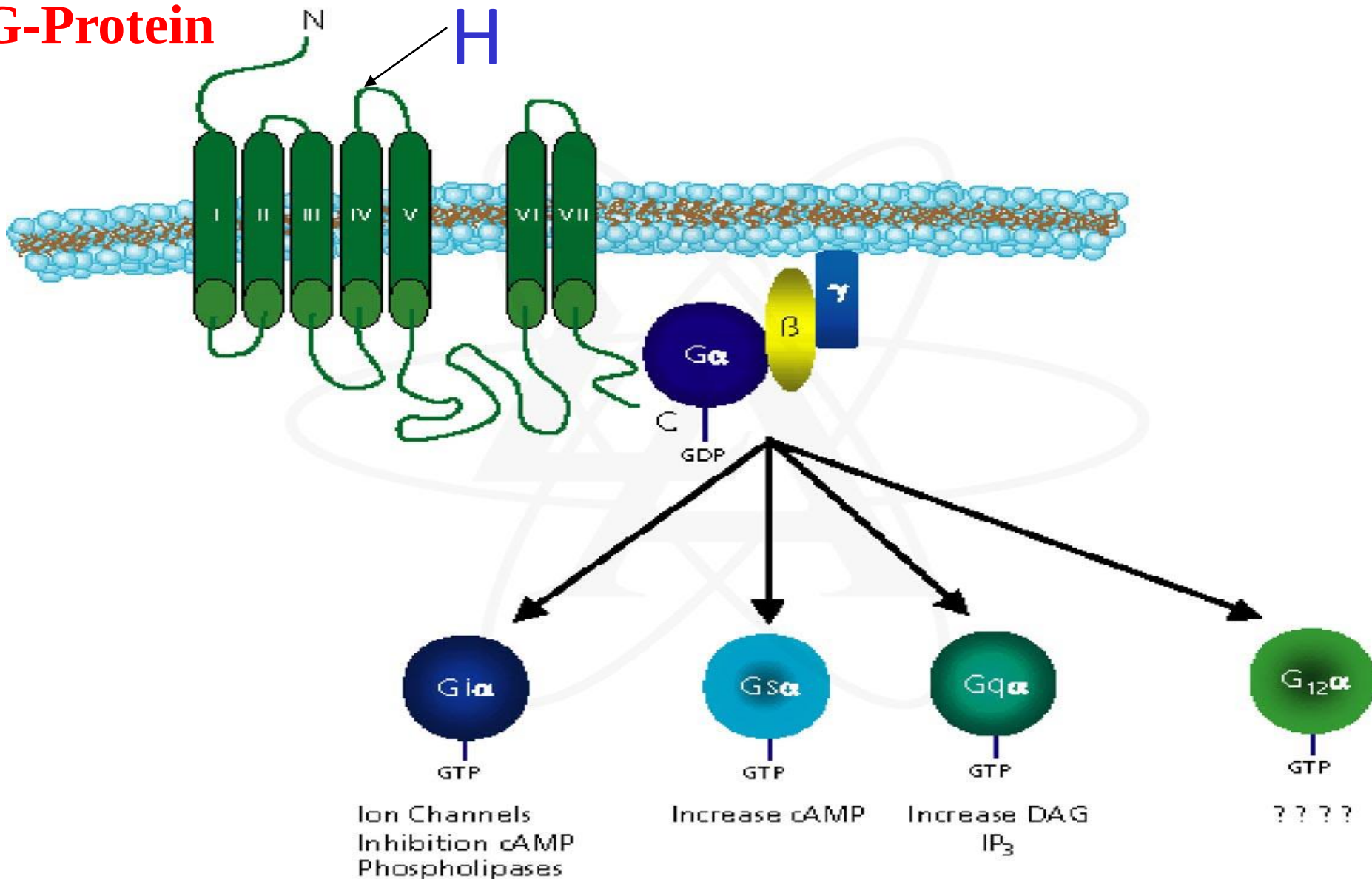


Hormones: MoA

2. Hydrophilic hormones:

- Catecholamines, Polypeptides/Proteins, Glycoproteins
- Since they can't easily cross the plasma membrane, they bind to cell surface receptors and activate/inhibit 2nd messengers
- G-protein coupled 2nd messengers
 - Adenylate cyclase - cyclic AMP (AC-cAMP)
 - Guanylate cyclase - cyclic GMP (GC-cGMP)
 - Phospholipase-C - IP3/DAG
- Tyrosine kinase coupled 2nd messengers

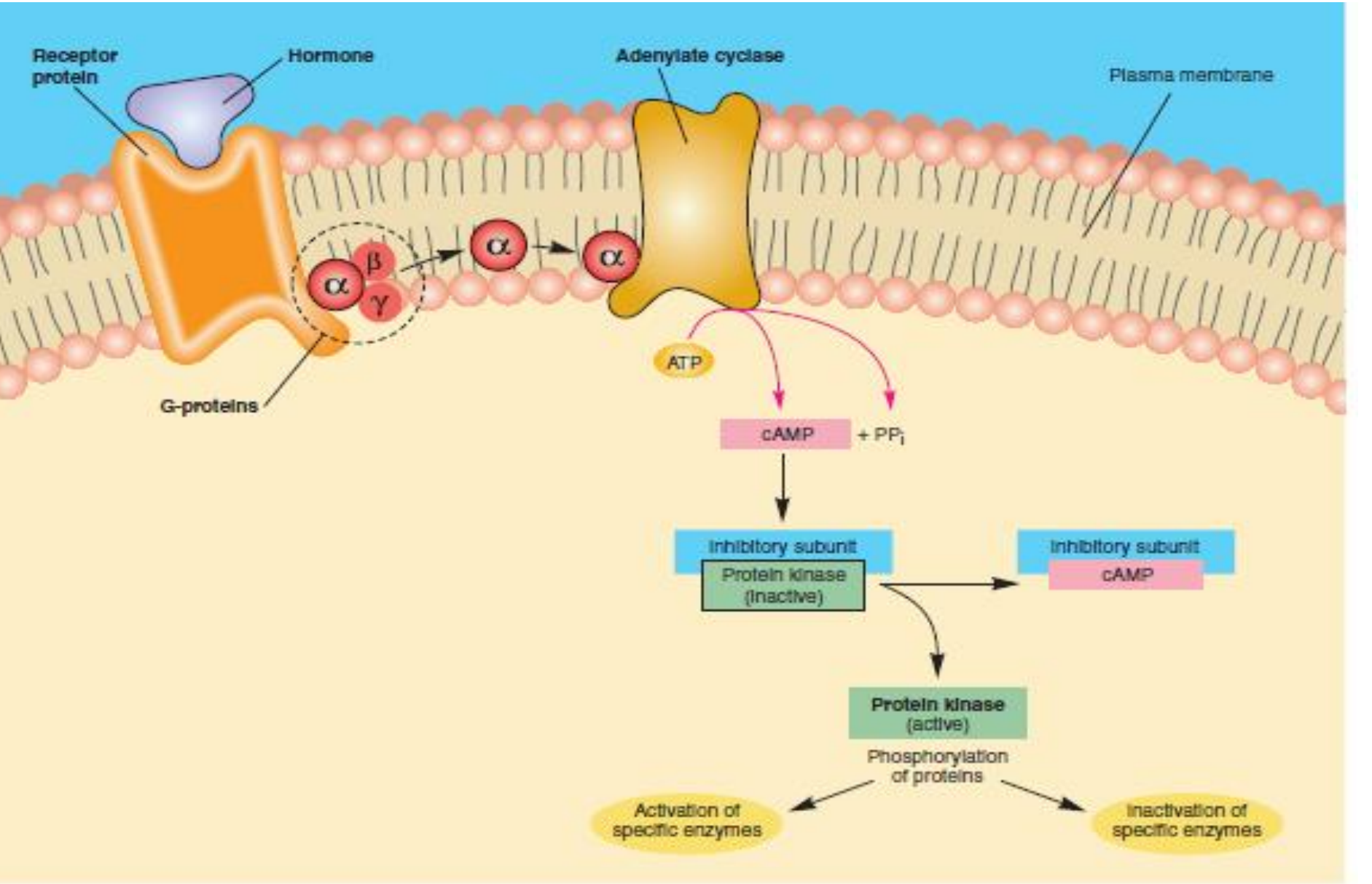
G-Protein

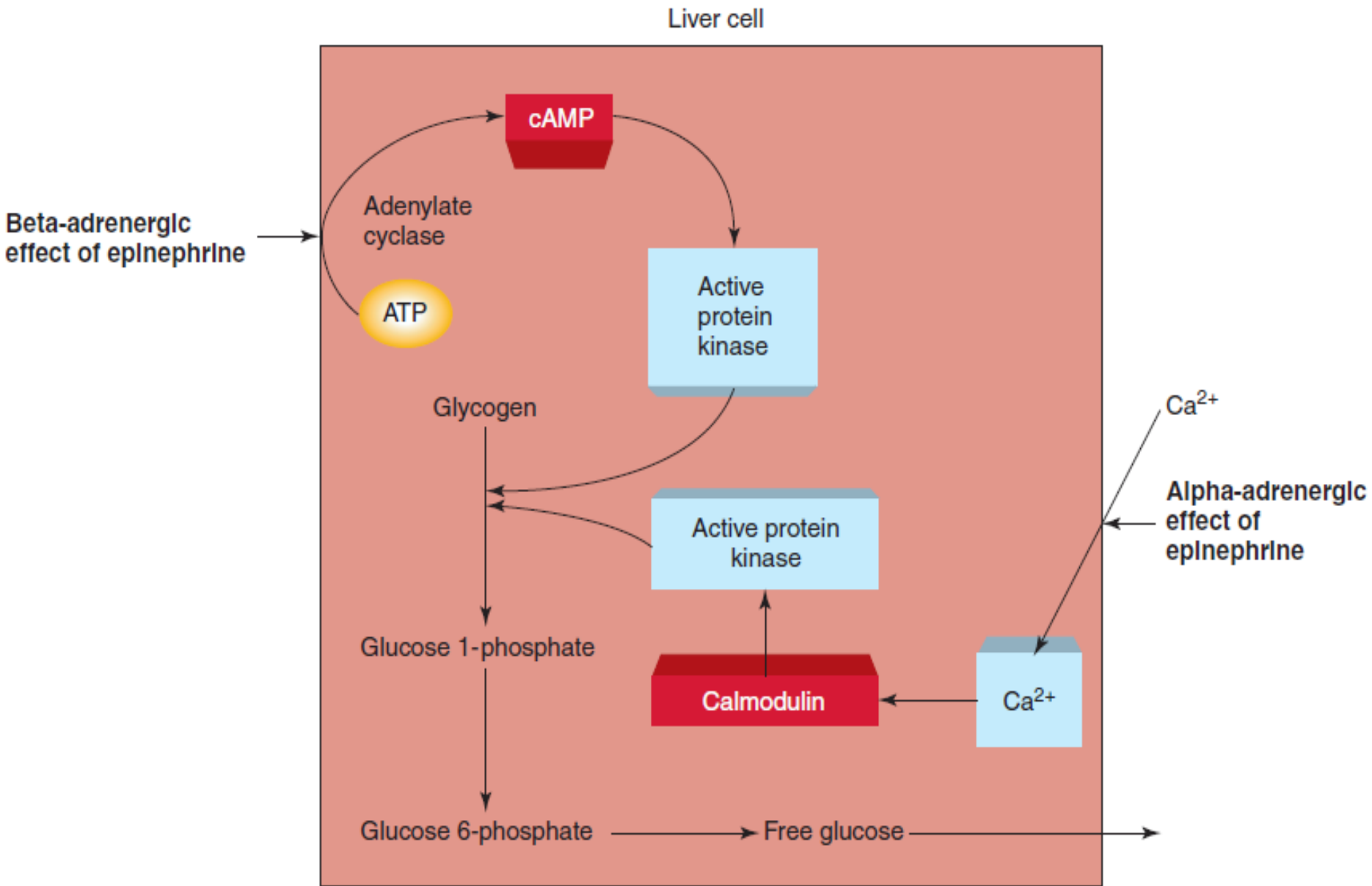


2nd Messengers

1. AC-cAMP

- When a hormone binds to G-protein coupled receptor, it causes dissociation of the $G\alpha$ -subunit from the G-protein complex ($G\alpha\beta\gamma$)
- The $G\alpha$ -subunit activates **Adenylate Cyclase (AC)**
- AC catalyzes conversion of: $ATP \rightarrow cAMP + P_{pi}$
- cAMP activates **protein kinase-A** (normally inactive) by binding to its inhibitory subunit
 - **Protein kinase** has two subunits: Catalytic + Inhibitory subunit
- Binding of cAMP to the inhibitory subunit causes it to dissociate from the catalytic subunit $\rightarrow$ **Protein kinase activation**





2nd Messengers

1. AC-cAMP

- Active protein kinase catalyzes the phosphorylation of different proteins in the target cells → enzyme activation/inactivation
- cAMP is inactivated in to 5'-AMP by **Phosphodiesterase enzyme (PDE)**

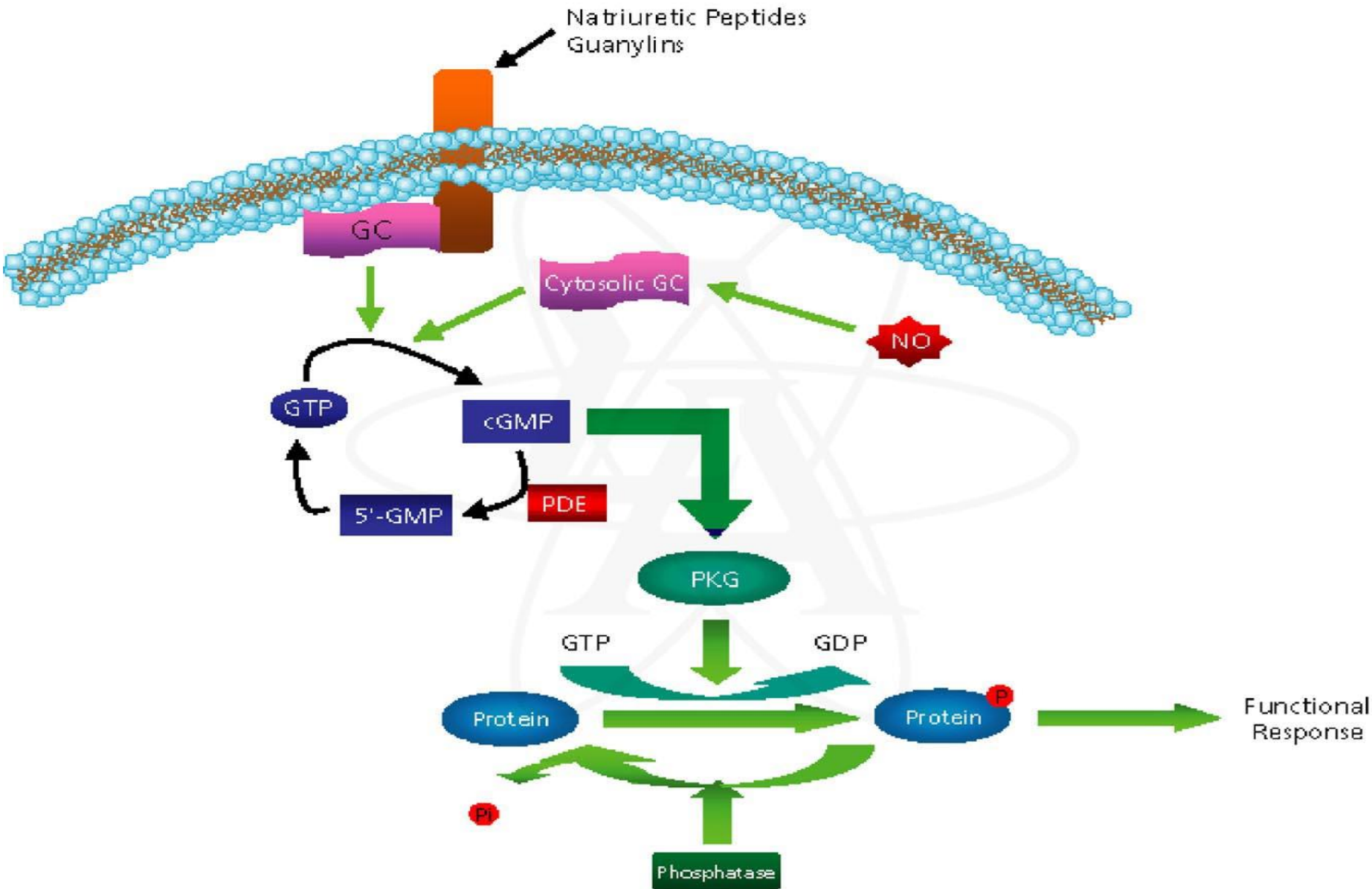
NB

- Drugs that inhibit PDE prevent the breakdown of cAMP and result in increased concentration of cAMP with in the target cell
- **Theophylline and Caffeine** → used clinically to raise cAMP levels and enhances the effect of Epinephrine (β_2) on bronchioles in Asthma patients

2nd Messengers

2. GC-cGMP

- The dissociated G α -subunit activates **Guanylate Cyclase (GC)**
- GC catalyzes conversion of: **GTP $\rightarrow$ cGMP + P_{pi}**
 - cGMP activates **protein kinase-G** $\rightarrow$ Phosphorylates cellular proteins $\rightarrow$ cellular response
 - cGMP is inactivated in to 5'-GMP by **Phosphodiesterase enzyme (PDE)**

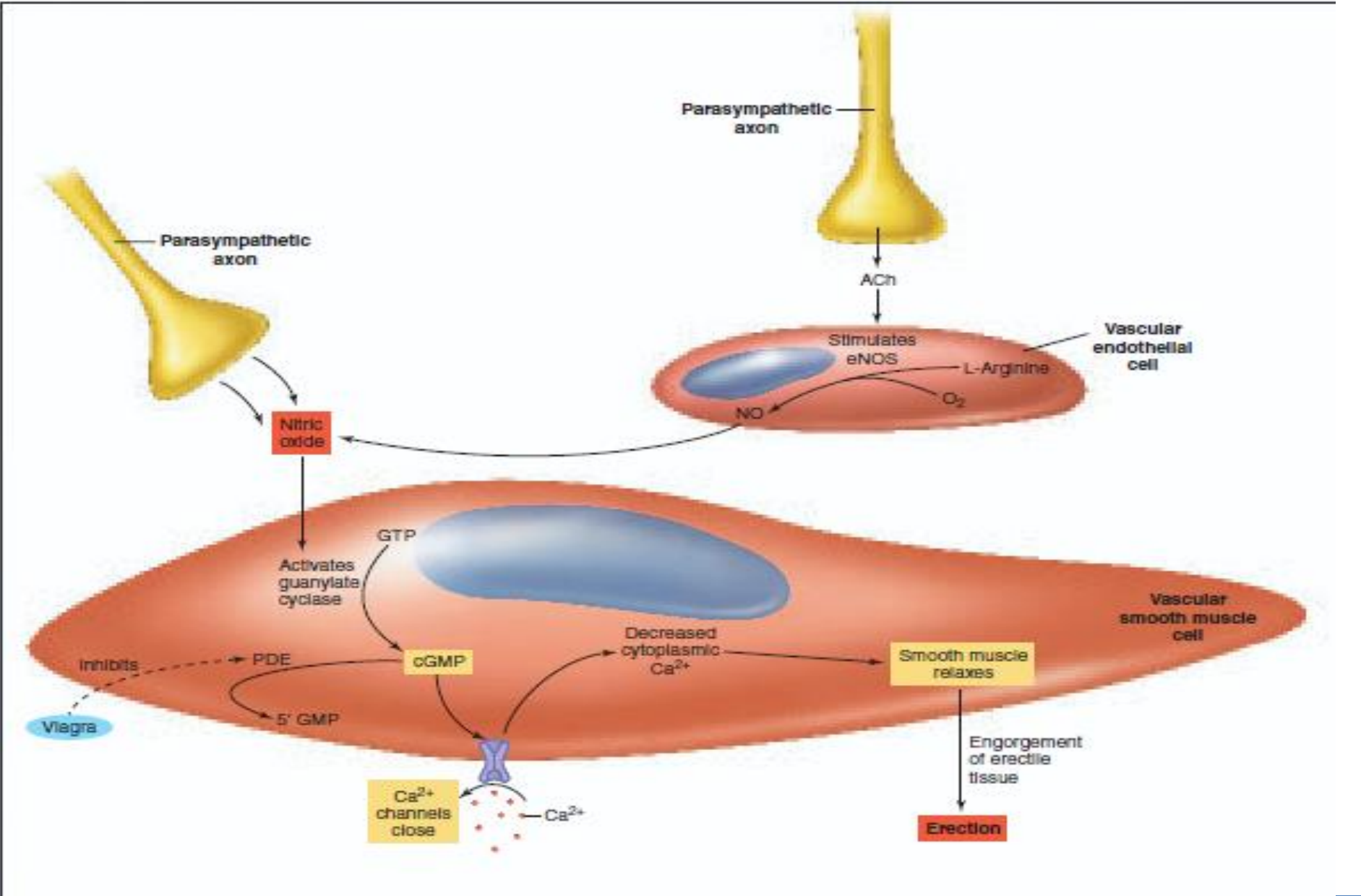


2nd Messengers

2. GC-cGMP

NB

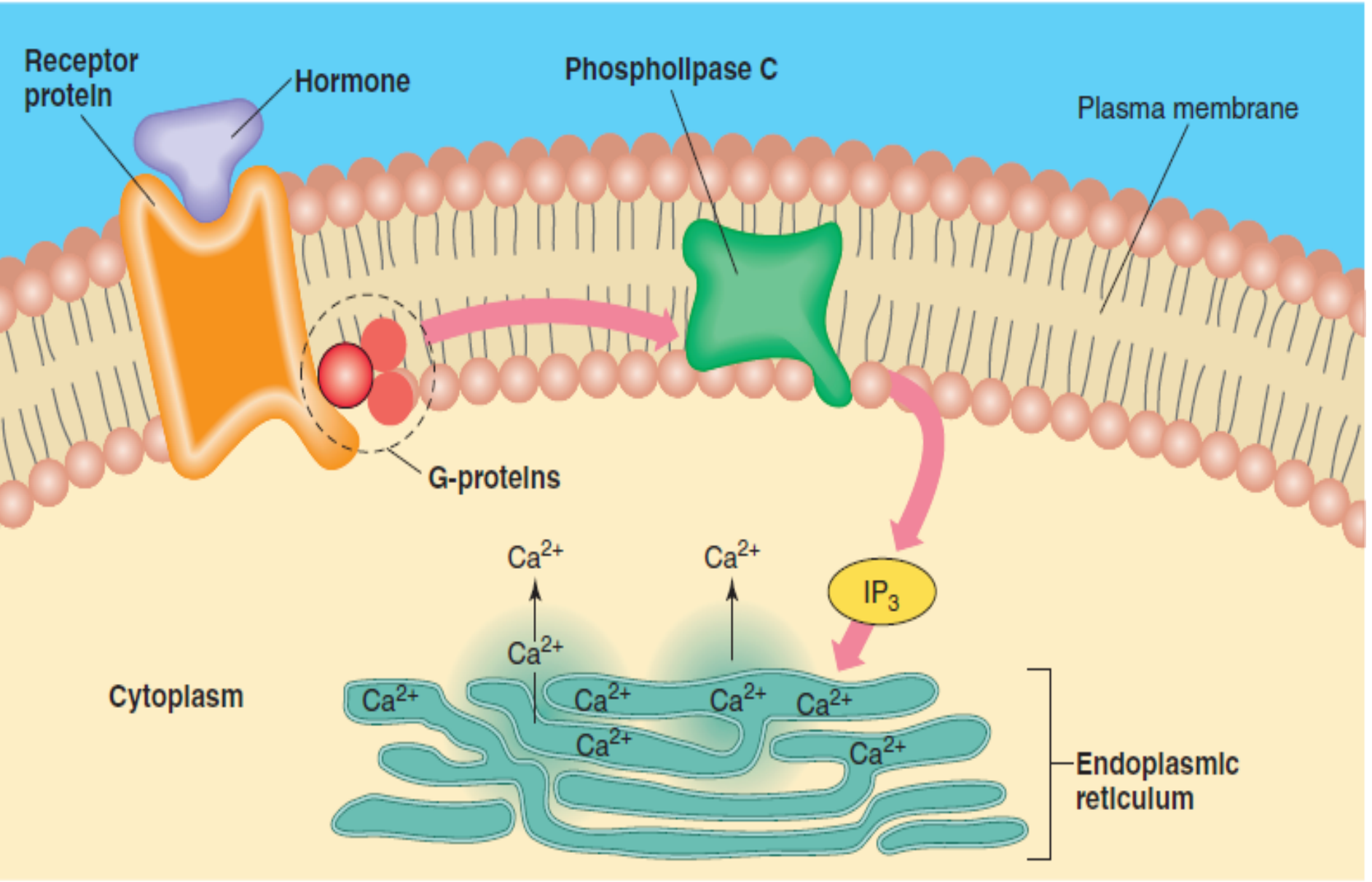
- Nitric oxide (NO) exert its dilator effect on smooth muscles by stimulating the production of cGMP in its target cells through activation of Guanylate Cyclase
 - Vascular smooth muscle relaxation that produces penile erection
 - **Viagra** helps treat erectile dysfunction by inhibiting PDE



2nd Messengers

3. PLC-IP3/DAG

- Binding of a hormone to its receptor activates phospholipase-C (PLC) via G-protein intermediate
- PLC catalyzes conversion of: **PIP2** → **IP3** + **DAG**
 - **IP3** binds to its receptors on the sarcoplasmic reticulum → open specific Ca^{2+} channels → Ca^{2+} diffuses into the cytoplasm → Ca^{2+} binds to **Calmodulin** → activates specific protein kinase enzymes



2nd Messengers

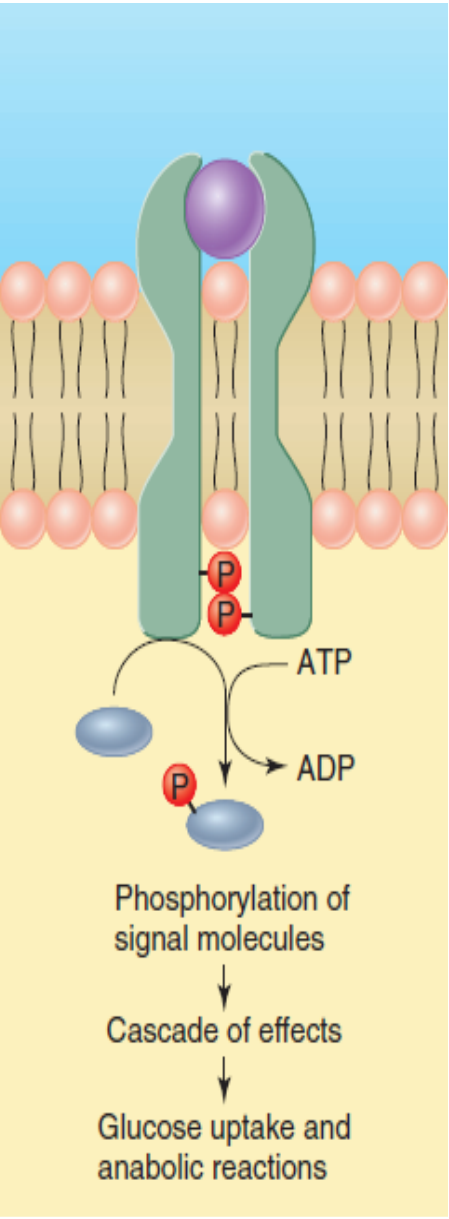
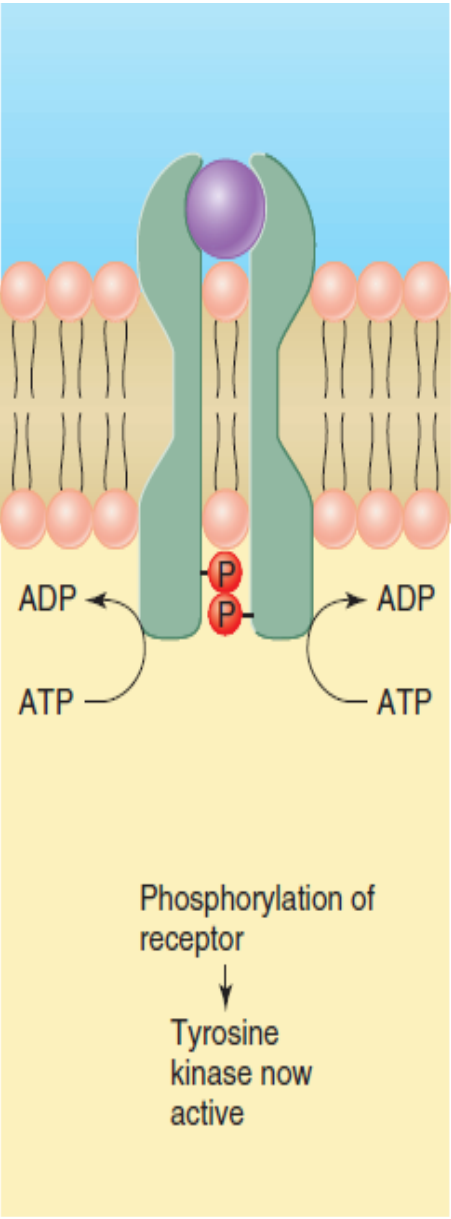
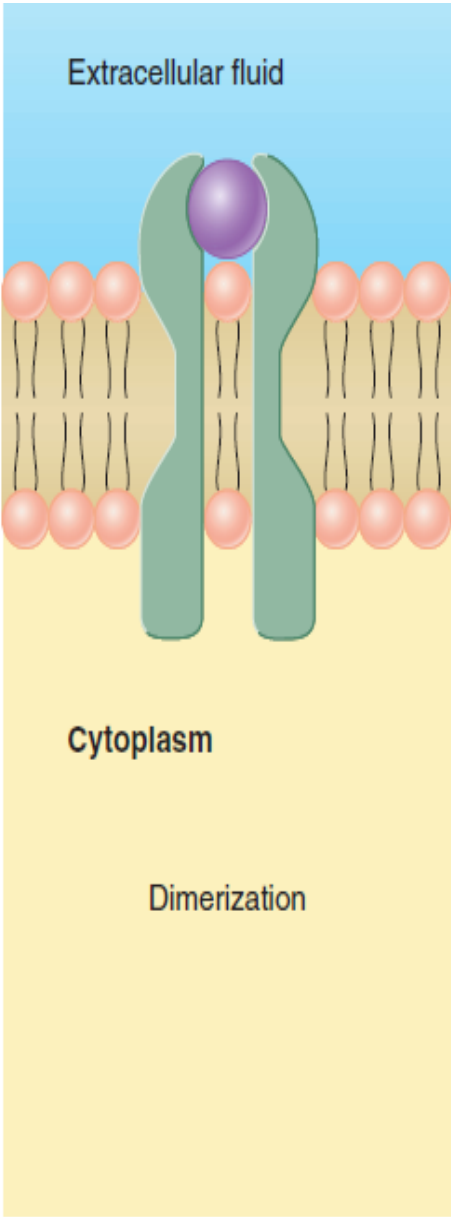
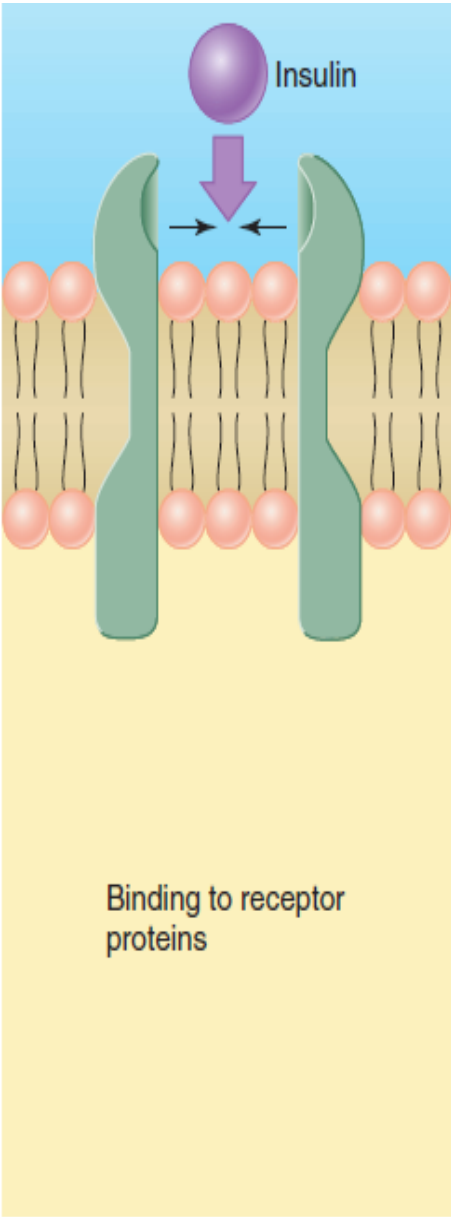
4. Tyrosine kinase coupled

- Tyrosine Kinase is located in the plasma membrane and serves both as a **receptor** and an **enzyme**
- Tyrosine kinase specifically phosphorylate tyrosine amino acid within the proteins
- **Insulin and Growth factors** (EGF, PDGF, IGFs) uses Tyrosine kinase 2nd messenger system
- **Insulin receptor (IR)** consists 2 subunits that dimerize when they bind with insulin

2nd Messengers

4. Tyrosine kinase coupled

- Each IR subunit contains: Insulin-binding site (outside) + enzymatic site (inside the cytoplasm)
 - The enzymatic site is inactive until insulin binds to the ligand-binding site and causes dimerization of the receptor
 - Insulin binding → receptor dimerization → enzymatic site activated in each unit of the receptor → one unit phosphorylates the other (*autophosphorylation*) → tyrosine kinase become active → phosphorylation of proteins (*signaling molecules*) → Response



2nd Messengers

4. Tyrosine kinase coupled

- Insulin stimulates **plasma glucose uptake** into the plasma membrane of skeletal muscle, adipose, and liver cells indirectly by stimulating the insertion of GLUT-4 carrier proteins (for facilitated diffusion of glucose)
- Insulin catalyzes **Glycogen production** in the liver and skeletal muscles indirectly by activation of **glycogen synthetase**